

Applied Statistics and Machine Learning

Customer Support Ticket Resolution Time using Linear Regression and SVR

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Lecturer	Assem Abdel-Hak
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Group Members	Varsha Sundararaj-20065968 Nandhitha Ganesan Rajambal-20069568

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1. Introduction

Customer support is a crucial factor that facilitates customer satisfaction and business operations in contemporary business environments. The primary organizational issue is to determine how long it will take to solve customer issues. When companies have the correct prediction of the time to resolve tickets, they can use this to improve their distribution of resources and response mechanisms, and their performance in service delivery.

The project uses two regression models: Linear Regression and Support Vector Regression (SVR) to forecast the duration of customer support tickets to resolve. Linear Regression is used as a baseline model and SVR is used to deal with non-linear, complex data patterns. Regression performance measures are used to test the models.

2. Dataset Description

2.1 Dataset Source

Kaggle (<https://www.kaggle.com/datasets/suraj520/customer-support-ticket-dataset>) was the source of Customer Support Ticket Dataset. The datasets include real operations information of customer services departments that can be utilized to carry out regression analysis.

2.2 Dataset Overview

The dataset has 8,469 records and 17 different columns that describe different aspects of customer support interactions. The dataset was preprocessed and feature engineered to include 67 features, and thus it was more appropriate to machine learning. The data set contains categorical as well as numeric variables, which allows the models to find patterns in different types of data.

2.3 Target Variable

The target variable is Resolution Hours - the total amount of time it takes to close a customer support ticket, defined as Time to Resolution - time first response. The target variable is a continuous variable and hence, this analysis is performed by regression.

3. Data Preparation Pipeline

3.1 Data Cleaning

The cleaning of the data is an important procedure since machine learning models require full data, free of errors. The dataset initially had multiple blanks in the columns like Resolution, First Response Time, Time to Resolution and Customer Satisfaction Rating.

The mode method (most frequent value per column) was used to fill in missing values. The unnecessary columns were then eliminated which are Ticket ID, Customer Name, Customer Email, Ticket Subject, Ticket Description, and Resolution as they are not relevant to prediction.

3.2 Data Encoding

Machine learning algorithms require numeric input. One-Hot Encoding was used to transform categorical variables, based on the following features:

- Customer Gender
- Product Purchased
- Ticket Type
- Ticket Status
- Ticket Priority
- Ticket Channel

The encoding operation increased the number of features to 66 columns, which allowed the model to identify more complicated relationships among various variables.

3.3 Date Conversion and Feature Engineering.

Date and time columns initially were of object (text) type. These were transformed into date time format to allow analysis by time. A target variable Resolution Hours was then obtained by computing the difference between Time to Resolution and First Response Time.

The date columns were used to extract additional features:

- Purchase Month
- Purchase Day
- First Response Hour
- First Response Minute

The features allow the model to discover time-based trends within the data. With the incorporation of these columns, the number of features went up to 71.

3.4 Final Cleaning and Feature Selection.

Once the feature extraction was done, new numeric columns were examined on whether they had any missing values and replaced the values with the median, which is proper when dealing with continuous variables. The original datatype of all the datetime columns was subsequently removed, leaving the dataset of 68 numeric-only tables. The data were now clean and modelling was possible.

3.5 Data Scaling

All input features were transformed with StandardScaler, so that each feature was made to have a mean value of 0 and a standard deviation of 1. Scaling is necessary to SVR since it involves distance-based computations, otherwise scaled features with large absolute values would overshadow the model. The feature matrix X was the only variable that was scaled, whereas the target variable y was retained in its original unit (hours).

3.6 Train-Test Split

The scaled data were divided into 80 percent training and 20 percent testing by train test split of sklearn. This created training set of 6,775 records and testing set of 1,694 records. The training on a part and testing against unknown data provide a fair estimate of performance.

4. Model Implementation

4.1 Linear Regression

The OLS (Ordinary Least Squares) method of statsmodels library was used to implement Linear Regression. The best-fit line was found by minimising the total squared error between the predicted and actual values, `add_constant()` was used to add an intercept to the input data, a requirement of statsmodels OLS.

The regression equation will be the form:

$$\text{Resolution Hours} = 10.34 + (0.02 \times x_1) + (-0.03 \times x_2) + \dots + (-1.73 \times x_{67})$$

The OLS model had a R^2 of about 0.492 which implies that it accounts 49.2 percent of the variation in the time of resolution.

4.2 Support Vector Regression (SVR)

SVR is a non-linear relationship model that employs kernel functions to predict an input-target relationship that is complex and non-linear. Rather than reducing the overall error at all points, SVR fits the data in an epsilon tube, and only penalises any points that are outside the margin.

The project used the SVR with the RBF (Radial Basis Function) kernel to learn non-linear patterns. The Linear Regression baseline has a slightly lower R2 than the basic SVR model (kernel=rbf, C=10, epsilon=0.5) and a R2 of 51.35.

5. Hyperparameter Optimization with GridSearchCV.

The SVR model was systematically tuned by using GridSearchCV with all combinations of the following parameters being tested:

- Kernel: linear, poly, rbf, sigmoid.
- Regularization parameter C: 1, 10
- Epsilon (error tolerance): 0.1, 1.0

Each combination was assessed honestly (without accessing the test set) by three-fold cross-validation (cv=3). The best parameters found were:

- Kernel: RBF
- C: 10
- Epsilon: 0.1

Overall, the tuned SVR model had the highest R2 at 52.47, thus it is the best performing model.

6. Nature of the Data (Linear or Non-Linear)

The dataset is a combination of categorical and numerical variables and complicated interactions between numerous features. It is confirmed by the fact that the SVR with RBF kernel was more effective in comparison with the Linear Regression that non-linear patterns exist in the data. The dependence of the input features (ticket priority, channel, product type, response time components) on the resolution time is not directly linear and this is why SVR is a more appropriate method.

7. Overfitting Avoidance

When a model fits the training data too well including noise and cannot generalise to the new data, this is referred to as overfitting. To avoid overfitting the following methods were used:

- Train-test split: 80% training, 20% testing makes certain that the model is tested on unseen data.
- Cross-validation through GridSearchCV: 3-fold cross-validation was done to test each parameter combination without bias.
- Feature scaling: StandardScaler normalises features, to avoid dominance of high-magnitude features.
- Hyperparameter tuning: Best C and epsilon values will ensure that the model does not overly penalise minor errors.

The combination of these steps guarantees that the model extrapolates much beyond the training data.

8. Model Evaluation

The main measure of evaluation of the three models was R2 (coefficient of determination). Results summary:

Model R 2 Score Notes

Linear Regression (OLS) 49.2% OLS baseline model.

SVR Method 1 (Manual RBF) 51.35% Better than LR

SVR Method 2 (GridSearchCV Tuned) 52.47% Best model

The tuned SVR model had the best score in R2. Although the R2 values are moderate, this is reasonable given real-world customer support datasets that have noise and variability built-in that cannot be represented by features.

9. Use of New Data to Predict.

In order to illustrate practical application, a sample row was chosen in the input data and with the help of all three trained models, predictions were made. The new input was scaled with the same StandardScaler that was trained on the training data, so as to be consistent.

Estimated response times of the sample input:

- Linear Regression: an approximation of 7.71 hours.
- Basic SVR: about 5.95 hours.
- Tuned SVR: around 6.15 hours.

These predictions show that the model predicts a resolution time of new and unknown customer support tickets.

10. Discussion

These findings show that good predictive performance is formed through appropriate model selection and deep data preprocessing.

- The quality of information to be used by the models was enhanced through data cleaning and encoding.
- The feature engineering (extracting date-based features, Resolution Hours) added meaningful patterns that the models could be learning.
- SVR performed better than Linear Regression because it was able to capture the non-linear relationships in the data.
- GridSearchCV further enhanced the accuracy of SVR by 51.35 to 52.47.

One limitation of this study is that the moderate values of R^2 indicate that some variables affecting the resolution time are not included in the data set, i.e., experience level of the agent, type of issue, or even time-of-day staffing.

11. Conclusion

This project has managed to analyze the Customer Support Ticket data with the help of a structured data preparation pipeline, and then trained and tested two regression models: Linear Regression and Support Vector Regression.

Key findings:

- The data presented has non-linear characteristics and thus SVR is the more suitable model.
- Best performance tuned SVR model (RBF with $C=10$, $\epsilon=0.1$) was found to have R^2 of 52.47%.
- Hyperparameter optimization using GridSearchCV performed better than the baseline models.
- The model proved to be able to forecast new customer support ticket resolution time.

Further research on the topic of future work might include the exploration of further features, ensemble models (Random Forest or Gradient Boosting), or deep learning-based methods to enhance the accuracy of prediction.

12. Individual Contributions

Student Name	What We Did
Nandhitha Ganesan Rajambal	Handled the full data preparation pipeline, including cleaning, encoding, and scaling. Implemented the Linear Regression model and wrote up the corresponding analysis and interpretation sections.
Varsha Sundararaj	Built and evaluated the SVR model, carried out hyperparameter tuning using GridSearchCV, and completed the prediction section. Also contributed to writing the overall report structure and the model evaluation discussion.